

Lab 4: VibeCoding — Chat History & Implementation Log

Student: Rohan Chukkapalli (SRN: PES1UG24CS135) | **Course:** Software Engineering Lab

Repository: SETAPESU26/09_tic_tac_toe | **Personal Repo:** VortexVector/09_tic_tac_toe

Tool: Google Antigravity AI (Pair Programmer)

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Assigned Repository: https://github.com/SETAPESU26/09_tic_tac_toe

Uploaded Document: Lab 4: VibeCoding Instructions.

Objective: Use Vibe Coding tools to fix broken code and add features in under 3-4 prompt iterations. Record 10s video before changes, fix win/draw detection, implement persistent scoreboard, add move validation, add first-player choice, record 10s video after changes, and push to personal repo without raising PR to SETAPESU26.

■ Antigravity AI (Pair Programming Assistant)

Repository analyzed and cloned into local workspace. Starter code preserved in tic-tac-toe-before/ for bug demonstration recording. Pygame environment configured on Python 3.13.

Prompt 1 (Task 1: Win/Draw Detection):

- Added diagonal lines [board[0][0], board[1][1], board[2][2]] and [board[0][2], board[1][1], board[2][0]] to rules.py.
- Prioritized check_winner() before is_board_full() in game_engine.py so winning moves on full boards are awarded properly instead of reporting Draw.
- Added if self.round_over: return in handle_click() to freeze input once a round ends.

Prompt 2 (Task 3: Turn & Move Validation):

- In handle_click(), validated cell state: if self.board[row][col] is not None: return to prevent overwriting existing symbols.

Prompt 3 (Tasks 2 & 4: Persistent Scoreboard & Starter Choice):

- Added persistent self.scores dictionary across rounds, new_round() (key [R]), reset_match() (key [M]), and toggle_starter() (key [T]).
- Automated AI first move when O is selected as starter.
- Updated renderer.py to draw turn status, scoreboard, result banners, and control reminders.

Unit Test Suite Execution:

Ran 7 tests covering all 4 tasks: test_diagonal_wins, test_win_on_last_move_is_not_draw, test_genuine_draw, test_occupied_cell_move_rejected, test_no_moves_accepted_after_round_over, test_persistent_scoreboard, test_starting_player_choice_and_computer_first_move.

Result: Ran 7 tests in 0.001s — OK (All passed).

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dont ask me always for permission always do wt is needed

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Acknowledged full autonomy. Automatically generated deliverable documents (PDF and DOCX chat history), structured the Lab-4/ folder containing Updated_Code/ and Videos/, and committed all changes to git (commit 6460ef4).

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is it done

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Confirmed all code changes, test verifications, and report generations are 100% complete. Outlined the steps to capture the 10s before and after screen recordings and push to personal GitHub.

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[Uploaded screenshot running tic-tac-toe-before/main.py]

Screen shows 3 X's on the anti-diagonal resulting in **'Draw! Press R for a new round'** and **'Computer's turn (O)'**.

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Confirmed screenshot clearly captures the starter bugs: diagonal win ignored and full-board draw shadowing the winner. Directed user to run tic-tac-toe/main.py to record the 10-second 'After' video demonstrating diagonal detection, move rejection, scoreboard updates, and starter toggle.

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k wt after taking video

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Instructed to place recordings in Lab-4/Videos/, create personal GitHub repo VortexVector/09_tic_tac_toe, and push with git push -u origin main without raising PR to SETAPESU26.

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wt nxt

[Uploaded screenshot showing successful push to https://github.com/VortexVector/09_tic_tac_toe.git]

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Detected uploaded video files (se_tictac_before.mp4 and se_tictac.mp4) on GitHub. Fetched, rebased, and pushed updated chat history exports (PDF, DOCX, MD) so both local and remote repositories are fully synced.

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gimme chat history

■ Antigravity AI (Pair Programming Assistant)

Provided direct file paths and complete transcript for the chat history deliverable in PDF, Word (DOCX), and Markdown formats.

■ User (Student: Rohan Chukkapalli)

take all permissions by default

■ Antigravity AI (Pair Programming Assistant)

Permissions accepted by default. Verified working directory is clean and fully synced with origin/main.

■ User (Student: Rohan Chukkapalli)

make a pdf of chat history

■ Antigravity AI (Pair Programming Assistant)

Generating professional, publication-quality PDF containing complete dialogue, prompt log, code diffs, and verification summary.